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Course L.A
Assignment 2

Ex 1.5

12) b) Inverse

$$\left[\begin{array}{ccc|ccc} \frac{1}{5} & \frac{1}{5} & \frac{-2}{5} & 1 & 0 & 0 \\ \frac{2}{5} & -\frac{3}{5} & \frac{-3}{10} & 0 & 1 & 0 \\ \frac{1}{5} & \frac{-4}{5} & \frac{1}{10} & 0 & 0 & 1 \end{array}\right]$$

Sol:-

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 2 & -3 & -\frac{3}{2} & 0 & 5 & 0 \\ 1 & -4 & \frac{1}{2} & 0 & 0 & 5 \end{array}\right] \quad R_1, R_2, R_3 \times 5$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & \frac{5}{2} & 10 & 5 & 0 \\ 0 & -5 & \frac{5}{2} & -5 & 0 & 5 \end{array}\right] \quad \begin{array}{l} -2R_1 + \text{add to } R_2 \\ -R_1 \text{ add to } R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & \frac{5}{2} & 10 & 5 & 0 \\ 0 & 0 & 0 & -15 & -5 & 0 \end{array}\right] \quad -R_2 \text{ add to } R_3$$

Inverse not possible because 0 row exists.

15 $\left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 2 & 7 & 7 & 0 & 0 & 1 \end{array}\right]$

$\left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array}\right]$ $-R_1$ add to R_2, R_3

$\left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$ $-R_2$ add to R_3

$\left[\begin{array}{ccc|ccc} 2 & 6 & 0 & 1 & 6 & -6 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$ $-6R_3$ add to R_1

$\rightarrow \frac{1}{2}$

$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & \frac{1}{2} & 3 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$ $R_1 \times \frac{1}{2}$

$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & 0 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$ $-3R_2$ add to R_1

Ans.

26 $\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$

$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$

E_1

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array}\right]$$ Exchange R_2, R_3

E_2

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array}\right]$$ $-R_3$ add to R_2

E_3

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array}\right]$$ $-R_2$ add to R_1

E_4

Since $E_4E_3E_2E_1A = I$

$A = (E_4E_3E_2E_1)^{-1} \cdot I$

$A = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A^{-1} = E_4E_3E_2E_1 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

23) $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 9 & 4 \\ -5 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$

$$\left[\begin{array}{ccc} 2 & 1 & 0 \\ -5 & -1 & 0 \\ 3 & 0 & -1 \end{array}\right]$$ $-2R_1$ add to R_2

$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 & 0 \\ -5 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix} \quad -2R_1 \text{ add to } R_3$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 9 & 4 \\ -5 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix} \quad -4R_3 \text{ add to } R_1$$

$$E_3 = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

since $E_3 E_2 E_1 A = B \qquad CA = B$

$$C = E_3 E_2 E_1 = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ex 1.6

4) $5x_1 + 3x_2 + 2x_3 = 4$

$3x_1 + 3x_2 + 2x_3 = 2$

$x_2 + x_3 = 5$

$Ax = b$

$$A = \begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$x = A^{-1}b$

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$$\left[\begin{array}{ccc|ccc} 5 & 3 & 2 & 1 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \quad -R_2 \text{ add to } R_1, \quad \frac{R_1}{2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 3 & 2 & -3/2 & 5/2 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \quad -3R_1 \text{ add } R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 3 & 2 & -3/2 & 5/2 & 0 \end{array}\right] \quad R_2 \text{ exchange } R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -3/2 & -5/2 & 3 \end{array}\right] \quad -3R_2 \text{ add } R_3, \quad -1R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 0 & -3/2 & 5/2 & -2 \\ 0 & 0 & 1 & 3/2 & -5/2 & 3 \end{array}\right] \quad -R_3 \text{ add } R_2$$

$$x = A^{-1}b = \begin{bmatrix} 1/2 & -1/2 & 0 \\ -3/2 & 5/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{pmatrix} 2 & -1 & +0 \\ -6 & +5 & +0 \\ 6 & -5 & 15 \end{pmatrix} = \begin{pmatrix} 1 \\ -11 \\ 16 \end{pmatrix} \quad \begin{matrix} x_1 = 1 \\ x_2 = -11 \\ x_3 = 16 \end{matrix}$$ Ans.

12) $x_1 + 3x_2 + 5x_3 = b_1$

$-x_1 - 2x_2 = b_2$

$2x_1 - 5x_2 + 4x_3 = b_3$

i) $b_1 = 1, \; b_2 = 0, \; b_3 = -1$

ii) $b_1 = 0, \; b_2 = 1, \; b_3 = 1$

iii) $b_1 = -1, \; b_2 = -1, \; b_3 = 0$

$$\left(\begin{array}{ccc|c|c|c} 1 & 3 & 5 & 1 & 0 & -1 \\ -1 & -2 & 0 & 0 & 1 & -1 \\ 2 & 5 & 4 & -1 & 1 & 0 \end{array}\right)$$

$$= \left(\begin{array}{ccc|c|c|c} 1 & 3 & 5 & 1 & 0 & -1 \\ 0 & 1 & 5 & 1 & 1 & -2 \\ 0 & -1 & -6 & -3 & 1 & +2 \end{array}\right) \begin{matrix} R_2 + R_1 \\ -2R_1 \text{ add } R_3 \end{matrix}$$

$$= \left(\begin{array}{ccc|c|c|c} 1 & 3 & 5 & 1 & 0 & -1 \\ 0 & 1 & 5 & 1 & 1 & -2 \\ 0 & 0 & +1 & +2 & -2 & 0 \end{array}\right) \begin{matrix} R_3 + R_1 \\ -1R_3 \end{matrix}$$

$$= \left(\begin{array}{ccc|c|c|c} 1 & 3 & 0 & -9 & -10 & -1 \\ 0 & 1 & 0 & -9 & 11 & -2 \\ 0 & 0 & 1 & 2 & -2 & 0 \end{array}\right) \quad -5R_3 \text{ add to } R_1, R_2$$

$$\left(\begin{array}{ccc|c|ccc|c} 1 & 0 & 0 & 18 & -23 & | & 5 \\ 0 & 1 & 0 & -9 & 11 & | & -2 \\ 0 & 0 & 1 & 2 & -2 & | & 0 \end{array}\right) \quad -3R_2 \text{ add to } R_1$$

SOLUTIONS are

i) $x_1 = 18, \quad x_2 = -9, \quad x_3 = 2$

ii) $x_1 = -23, \quad x_2 = 11, \quad x_3 = -2$

iii) $x_1 = 5; \quad x_2 = -2, \quad x_3 = 0$

b) $x_1 - x_2 + 3x_3 + 2x_4 = b_1$

$-2x_1 + x_2 + 5x_3 + x_4 = b_2$

$-3x_1 + 2x_2 + 2x_3 - x_4 = b_3$

$4x_1 - 3x_2 + x_3 + 3x_4 = b_4$

$$= \left(\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ -2 & 1 & 5 & 1 & b_2 \\ -3 & 2 & 2 & -1 & b_3 \\ 4 & 3 & 1 & 3 & b_4 \end{array}\right)$$

$$= \left(\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ 0 & 1 & -11 & -5 & -2b_1 + b_2 \\ 0 & -1 & 11 & 5 & 3b_1 + b_3 \\ 0 & 1 & -11 & -5 & -4b_1 + b_4 \end{array}\right) \quad \begin{array}{l} R_2 + 2R_1 \\ R_3 + 3R_1 \\ R_4 + -4R_1 \end{array} \qquad \frac{R_2}{-1}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ 0 & 1 & -11 & -5 & -2b_1 + b_2 \\ 0 & 0 & 0 & 0 & b_1 - b_2 + b_3 \\ 0 & 0 & 0 & 0 & -2b_1 + b_2 + b_4 \end{array}\right) \quad \begin{array}{l} R_3 + R_2 \\ R_4 + -1R_2 \end{array}$$

System is consistent if

$b_1 - b_2 + b_3 = 0$ & $-2b_1 + b_2 + b_4 = 0$

so

$b_2 = b_1 + b_3$ & $b_2 + b_4 = 2b_1$

Ex 1.7

6) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 & 3 \\ 1 & 2 & 0 \\ -5 & 1 & -2 \end{bmatrix} \begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 6 & 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} (2)(4)(-3) & 2(-1)(5) & 2(3)(2) \\ (-1)(1)(-3) & (-1)(2)(5) & (-1)(0)(2) \\ (4)(-5)(-3) & (4)(1)(5) & (4)(-2)(2) \end{bmatrix}$

$= \begin{bmatrix} -24 & -10 & 12 \\ 3 & -10 & 0 \\ 60 & 20 & -16 \end{bmatrix}$

10) A^2, A^{-2}, A^{-k}

$\begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ $A^2 = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

$A^{-2} = \begin{pmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{16} & 0 & 0 \\ 0 & 0 & \frac{1}{9} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{pmatrix}$, $A^{-k} = \begin{pmatrix} \frac{1}{(-2)^k} & 0 & 0 & 0 \\ 0 & \frac{1}{(-4)^k} & 0 & 0 \\ 0 & 0 & \frac{1}{(-3)^k} & 0 \\ 0 & 0 & 0 & \frac{1}{2^k} \end{pmatrix}$

10) a) $\begin{pmatrix} u & v \\ w & x \\ y & z \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$

$= \begin{pmatrix} ua & vb \\ wa & xb \\ ya & zb \end{pmatrix}$

22) $\begin{pmatrix} 2 & 0 & 0 & 0 \\ -3 & -1 & 0 & 0 \\ -4 & -6 & 0 & 0 \\ 0 & 3 & 8 & -5 \end{pmatrix}$ For matrix to be invertible all diagnal entries are non-zero 3rd, 4th cloumn is 0

24) $A = \begin{pmatrix} 4 & 0 & 0 \\ -2 & 0 & 0 \\ -3 & 0 & 7 \end{pmatrix}$ $B = \begin{pmatrix} 6 & 0 & 0 \\ 1 & 5 & 0 \\ 3 & 2 & 6 \end{pmatrix}$

$= \begin{pmatrix} 4 \times 6 & 0 & 0 \\ -12 & 0.5 & 0 \\ 3 & 0.5 & 42 \end{pmatrix}$

26) $A = \begin{bmatrix} 2 & a-2b+2c & 2a-b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$

$a-2b+2c = 3$

$2a+b+c = 0$

$a+c = -2$

28) $A = \begin{bmatrix} x-\frac{1}{2} & 0 & 0 \\ 0 & x-\frac{1}{3} & 0 \\ 0 & 0 & x+\frac{1}{4} \end{bmatrix}$

matrix is invertible iff all diagnol entries are non-zero therefore

$x \neq \frac{1}{2}, x \neq \frac{1}{3}, x \neq -\frac{1}{4}$ otherwise

A is invertible.

32)

$A^2 = \begin{bmatrix} a & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $A = ?$

$A = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

also $A = \begin{bmatrix} \frac{-1}{3} & 0 & 0 \\ 0 & \frac{-1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and more.

Ex 1.8

4) b) $w_1 = 2x_1 + 2x_2 - 4x_3$

$w_2 = 4x_1 - 3x_2 + 2x_3$

Domain = 3

Codomain = 2 $\quad \mathbb{R}^3 \to \mathbb{R}^2$

6) b)

$$\begin{bmatrix} 2 & 1 & 6 \\ 3 & 7 & -4 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\mathbb{R}^3 \to \mathbb{R}^3$

Codomain = 3 $\quad$ Domain = 3

12) b) $w_1 = x_1$

$w_2 = x_1 + x_2$

$w_3 = x_1 + x_2 + x_3$

$w_4 = x_1 + x_2 + x_3 + x_4$

sol

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Matrix of transformation

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

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13) d) $T(x_1, x_3, x_3, x_4) = (x_4, x_1, x_3, x_2, x_1 - x_3)$

$$= \begin{bmatrix} x_4 \\ x_1 \\ x_3 \\ x_2 \\ x_1 - x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

standard matrix $= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix}$

16) $w_1 = 2x_1 + 3x_2 - 5x_3 - x_4$

$w_2 = x_1 - 5x_2 + 2x_3 - 3x_4$

standard matrix $= \begin{bmatrix} 2 & 3 & -5 & -1 \\ 1 & -5 & 2 & -3 \end{bmatrix}$

i) directly putting in equations $T(1, -1, 2, 4)$

$w_1 = 2(1) + 3(-1) - 5(2) - 4$

$w_2 = 1 - 5(-1) + 2(2) - 3(4)$

$w_1 = -15$

$w_2 = -2$

ii) by matrix multiplication

$$\begin{bmatrix} 2 & 3 & -5 & -1 \\ 1 & -5 & 2 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 - 3 - 10 - 4 \\ 1 + 5 + 4 - 12 \end{bmatrix} = \begin{bmatrix} -15 \\ -2 \end{bmatrix}$$

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22) a) $T(x,y,z) = (x+y, y+z, x)$

let $u = u_1, u_2, u_3$

$v = v_1, v_2, v_3$

$T(u+v) = T(u_1+v_1, u_2+v_2, u_3+v_3)$

$= (u_1+v_1+u_2+v_2, u_2+v_2+u_3+v_3, u_1+v_1)$

$T(u) = u_1+u_2, u_2+u_3, u_1$

$T(v) = v_1+v_2, v_2+v_3, v_1$

$T(u)+T(v) = (u_1+u_2+v_1+v_2, v_2+v_3+u_2+u_3, u_1+v_1)$

$T(ku) = (ku_1+ku_2, ku_2+ku_3, ku_1)$

$= k(u_1+u_2, u_2+u_3, u_1)$

hence it is transformation.

b) $T(x_1, x_2) = x_2, x_1$

let $u = u_1, u_2$

$v = v_1, v_2$

$T(u+v) = (v_2+u_2, v_1+u_1)$

$= (u_2, u_1) + (v_2, v_1)$

$= T(u) + T(v)$

$T(kv) = (kv_2, kv_1)$

$= k(v_2, v_1)$

24) a) $T(x,y) = (x, y+1)$

$T(0) = (0,0)$ but

$T(0) = (0, 0+1) = (0, 1)$

b) $T(x_1, x_2, x_3) = (x_1, x_2, \sqrt{x_3})$ $u = u_1 u_2, u_3$

$T(ku) = kT(u)$

$= (ku_1, ku_2, \sqrt{ku_3}) \neq (ku_1, ku_2, k\sqrt{u_3})$

33) (+2, -5, 3) reflection

a) xy plane

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ -3 \end{bmatrix}$$

b) xz-plane

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

c) yz plane

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$$

35) orthogonal projection (-2, 1, 3)

a) xy plane

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

b) xz plane

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

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c) yz plane

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$$

32) c) image of $(3,-4)$ $\theta = 45°$

sol

$$= \begin{bmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{bmatrix} \begin{bmatrix} 3 \\ -4 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} - \frac{4}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 7/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

41) $R^3 \to R^3$ $A = \begin{bmatrix} -1 & 3 & 0 \\ 2 & 1 & 2 \\ 4 & 5 & -3 \end{bmatrix}$

a) $T_A(e_1) = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$

$T_A(e_2) = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}$

$T_A(e_3) = \begin{bmatrix} 0 \\ 2 \\ -3 \end{bmatrix}$

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b) $T_A(e_1+e_2+e_3)$

$$= \begin{bmatrix} -1+3+0 \\ 2+1+2 \\ 4+5-3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

c) $T_A(7e_3)$

$$= \begin{bmatrix} 0 \\ 14 \\ -21 \end{bmatrix}$$

45) $R^2 \to R^2$

$$T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

We need to find A

$T(x) = A(x)$

Let $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$T(v_1) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $T(v_2) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

Let $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

i) $\Rightarrow c_1 + 2c_2 = 1$

ii) $\Rightarrow c_1 + 3c_2 = 0$

iii) $\Rightarrow c_1 = 1 - 2c_2$ put in (ii)

$1 - 2c_2 + 3c_2 = 0$

$c_2 = -1$ put in (iii)

$c_1 = 1 - 2(-1)$

$c_1 = 3$

$e_1 = 3v_1 - v_2$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$c_1 + 2c_2 = 0$ (i)

$c_1 + 3c_2 = 1$ (ii)

$c_1 = -2c_2$ (iii) put in (ii)

$-2c_2 + 3c_2 = 1$

$c_2 = 1$ put in (iii)

$c_1 = -2(1) = -2$

$e_2 = -2v_1 + v_2$

$$T(e_1) = 3T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) - T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$$

$$= 3\begin{bmatrix} 1 \\ -2 \end{bmatrix} - \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3+2 \\ -6-5 \end{bmatrix} = \begin{bmatrix} 5 \\ -11 \end{bmatrix}$$

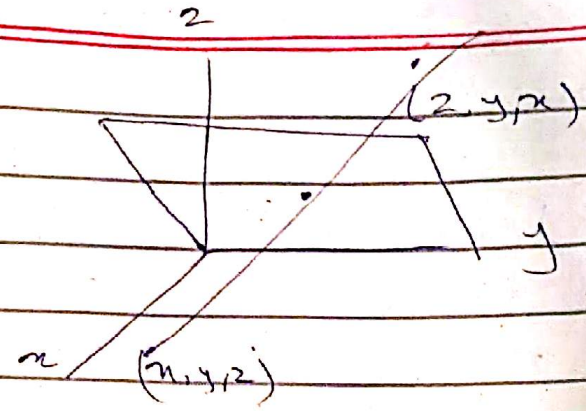
$$T(e_2) = -2T(v_1) + T(v_2)$$

$$= -2\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} -2+2 \\ +4+5 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \end{bmatrix}$$

$$A = [T(e_1) \quad T(e_2)] = \begin{bmatrix} 5 & -4 \\ -11 & 9 \end{bmatrix}$$ Ans.

48)b)



$$= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Ex 1.9

Example 3

$$T_{A_1} \circ T_{A_2} = T_{A_2} \circ T_{A_1}$$

$$T_{A_1} T_{A_2} = T_{A_2} T_{A_1}$$

$$A_1 = \begin{pmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{pmatrix} \qquad A_2 = \begin{pmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{pmatrix}$$

$$A_1 A_2 = \begin{pmatrix} \cos\theta_1\cos\theta_2 & -\sin\theta_1\sin\theta_2 & -\cos\theta_1\sin\theta_2 - \sin\theta_1\cos\theta_2 \\ \sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2 & -\sin\theta_1\sin\theta_2 + \cos\theta_1\cos\theta_2 & \end{pmatrix}$$

$$= \begin{pmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) \end{pmatrix}$$

$$A_2 A_1 = \begin{pmatrix} \cos(\theta_2+\theta_1) & -\sin(\theta_2+\theta_1) \\ \sin(\theta_2+\theta_1) & \cos(\theta_2+\theta_1) \end{pmatrix} = A_1 A_2$$

Example 7

$T: R^2 \to R^2$

$w_1 = 2x_1 + x_2$

$w_2 = 3x_1 + 4x_2$

$T^{-1}(w_1, w_2)$

We

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \qquad |A| = 8 - 3 = 5$$

$$A^{-1} = \frac{Adj\,A}{|A|} = \frac{1}{5}\begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & -\frac{1}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix}$$

$$A^{-1}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & -\frac{1}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{4}{5}w_1 & -\frac{1}{5}w_2 \\ -\frac{3}{5}w_1 & \frac{2}{5}w_2 \end{bmatrix}$$

From this we conclude that

$$T^{-1}(w_1, w_2) = \left(\frac{4}{5}w_1 - \frac{1}{5}w_2, -\frac{3}{5}w_1 + \frac{2}{5}w_2 \right)$$

4) $T_1: \mathbb{R}^3 \to \mathbb{R}^3$ reflection in xy plane

$T_2: \mathbb{R}^3 \to \mathbb{R}^3$ $T(x,y,z) = (2x, 3y, z)$

Sol.

$$T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad T_2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$[T_1 \circ T_2] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[T_2 \circ T_1] = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

So $T_1 \circ T_2 = T_2 \circ T_1$

8) a) $\theta = 60^\circ$ projection on x-axis reflection about line $y = x$

$T = T_3 \circ T_2 \circ T_1$

$$[T_1] = \begin{bmatrix} \sin 60 & -\cos 60 \\ \cos 60 & \sin 60 \end{bmatrix} \begin{bmatrix} \cos 60 & -\sin 60 \\ \sin 60 & \cos 60 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \quad [T_2] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$[T_3] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad T = [T_3][T_2][T_1]$$

$$T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \text{ Ans.}$$

b) $[T_1] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad [T_2] = \begin{bmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{bmatrix}$

$[T_3] = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad T = [T_3][T_2][T_1]$

$$T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \text{ Ans.}$$

c) $[T_1] = \begin{bmatrix} \cos 15 & -\sin 15 \\ \sin 15 & \cos 15 \end{bmatrix} \rightarrow [T_2]$

We know that multiple rotations have net effective addition of all rots.

$\theta = 15 + 105 + 60 = 180$

$$T = \begin{bmatrix} \cos(180) & -\sin(180) \\ \sin(180) & \cos(180) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

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10) a)

$$[T_1] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad [T_2] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$T_3(x) = -x$

$$T_3(x_1, x_2, x_3) = \begin{bmatrix} -x_1 \\ -x_2 \\ -x_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$[T_3] = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \qquad T = [T_3][T_2][T_1]$$

$$T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$ Ans.

b) $T = [T_3][T_2][T_1]$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

c) $T = (T_3)(T_2)(T_1)$

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$ Ans.

15) $T_1: R^2 \to R^4$ $T_2: R^4 \to R^3$

$T_1(x,y) = (y, x, x+y, x-y)$

$T_2(x,y,z,w) = (x+w, y+w, z+w)$

a) Standard matrix for T_1, T_2

in vector form $T(x,y) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

so

$$[T_1] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Vector form

$$T(x,y,z,w) = \begin{matrix} & x & y & z & w \\ \end{matrix}\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

$$(T_2) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

b) $[T_2 \circ T_1] = (T_2)(T_1)$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 2 & 0 \end{bmatrix}$$ Ans

c) $T_1 \circ T_2$ is not defined because outputs from T_2 are R^3 vectors but the inputs for T_1 are R^2 vectors

d) $T_2(T_1(x,y)) = (x, 2x-y, 2x)$ Ans.

$\frac{15}{2}+1$ $\frac{3}{2}+1$

18a) $w_1 = 2x_1 - 3x_2$

$w_2 = 5x_1 + x_2$

$$\begin{Bmatrix} w_1 \\ w_2 \end{Bmatrix} = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

$$A = \left[\begin{array}{cc|cc} 2 & -3 & 1 & 0 \\ 5 & 1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & -3/2 & \frac{1}{2} & 0 \\ 5 & 1 & 0 & 1 \end{array}\right] \quad \frac{R1}{2}$$

$$\left[\begin{array}{cc|cc} 1 & -3/2 & 1/2 & 0 \\ 0 & 19/2 & -\frac{5}{2} & 1 \end{array}\right] \quad -5R_1 \text{ add to } R_2$$

$$\left[\begin{array}{cc|cc} 1 & -3/2 & 1/2 & 0 \\ 0 & 1 & -\frac{5}{19} & 2/19 \end{array}\right] \quad R_2 / 19/2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 1/17 & 3/17 \\ 0 & 1 & -\frac{5}{17} & 2/17 \end{array}\right] \quad \frac{3R}{2} \text{ add to } R_1$$

19)a) $u_1 = x_1 + 2x_2$

$u_2 = -x_1 + x_2$

$$\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \quad |A| = 3 \neq 0$$

$$A^{-1} = \begin{bmatrix} 1/3 & 2/3 \\ 1/3 & 1/3 \end{bmatrix}$$

$$T^{-1}(w_1, w_2) = \left(\frac{1}{3}w_1 - \frac{2}{3}w_2,\ \frac{1}{3}w_1 + \frac{1}{3}w_2\right)$$

(4) a)

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix} \quad u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

inverse does not exist

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right]$$